Sanjana Sinha

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Education

Supaul College of Engineering, Bihar — Aug 2022 – May 2026
B.Tech. in Computer Science and Engineering — CGPA: 8.89

Gems English School — Aug 2020 – Jun 2022
ISC (Class XII) — 78.25%

Gems English School — Jun 2020
ICSE (Class X) — 85.6%

Experience

Software Development Intern — Dec 2025 – Feb 2026
Amnex Infotechnologies Pvt. Ltd., Ahmedabad

- Contributed to software development tasks, application module implementation, and debugging.
- Collaborated with developers to troubleshoot issues, test functionality, and follow coding practices.
- Gained hands-on experience in software development, backend concepts, and collaborative development workflows.

Projects

Lung Cancer Risk Prediction System
Python, Flask, Pandas, NumPy, Scikit-learn, PCA, SHAP, LIME
GitHub Repository

- Developed a machine learning web application using clinical and smoking-related features from the National Lung Screening Trial (NLST) dataset, comprising approximately 53,454 trial participants.
- Implemented Logistic Regression, Random Forest, and Naive Bayes models with missing-value imputation, feature scaling, PCA-based dimensionality reduction, and an 80:20 stratified train-test split.
- Evaluated models using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices.
- Integrated trained models into a Flask application with a weighted risk-scoring mechanism combining Logistic Regression (40%) and Random Forest (60%), and applied SHAP and LIME to explain model predictions.

Employee Management System
Angular, TypeScript, C#, ASP.NET Core Web API, EF Core, SQL Server, SignalR, Leaflet
GitHub Repository

- Developed a full-stack employee management application with role-based Admin and Employee interfaces for employee records, task assignments, attendance, and leave management.
- Engineered RESTful APIs using ASP.NET Core Web API, C#, Entity Framework Core, and SQL Server; integrated SignalR for real-time leave approval and rejection notifications.
- Implemented dashboard analytics, attendance history, and Leaflet-based employee location visualization with a 100-meter office geofence, using Git and GitHub for version control.

Technical Skills

Programming Languages: C++, Java, C#, Python
Web Development: HTML, CSS, JavaScript, TypeScript, Angular, Bootstrap
Backend: ASP.NET Core, .NET, Flask, RESTful APIs
Databases: SQL Server, MySQL, Entity Framework Core, SQL
Machine Learning: Logistic Regression, Random Forest, Naive Bayes, PCA, SHAP, LIME, Scikit-learn
Libraries and Tools: Pandas, NumPy, Matplotlib, Git, GitHub, VS Code, Jupyter Notebook, SignalR, Leaflet
Core Computer Science: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Achievements

- Cleared TCS CodeVita Season 13 Round 1, securing All India Rank (AIR) 1236.
- Won first prize in a Webathon (web development competition).
- Received a Certificate in Web Development from IIT Roorkee.